

Basin MCMC Assignment 1

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The purpose of this assignment is to break down the development of the initial basin probability distribution and probability in the Basin MCMC algorithm.

Question 1.

[pts.]

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_1$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_1 : \mathcal{X}_1 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. a good approximation to the log posterior in the region of high probability around θ_0 ;
3. easy to compute defining quantities from θ_0 ;
4. easy to sample from;
5. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. We present the solution as follows:

1. Define f_1 as the second-order Taylor polynomial of the log posterior about θ_0 ;
2. Derive a formula for the gradient $g = \nabla \log p(\theta_0 \mid y)$;
3. Derive a formula for the negative Hessian $P = -\nabla^2 \log p(\theta_0 \mid y)$;
4. State the final formula for f_1 .

1. For $\theta \in \mathcal{X}_1$, the log posterior is given by

$$\log p(\theta \mid y) = -\frac{1}{2\sigma^2} \|y - A(\theta)\|_2^2 + \log p(\theta) + C,$$

where C is some constant independent of θ , and $A : \mathcal{X} \rightarrow [0, 1]^{N_t N_x}$ is the forward map. Performing a second-order Taylor expansion about θ_0 (since $\log p(\theta \mid y) \in C^3(U)$ for some open neighbourhood $U \subset \mathcal{X}_1$ of θ_0 , provided $\theta_0 \in \text{int}(\mathcal{X}_1)$) gives

$$\log p(\theta \mid y) = \log p(\theta_0 \mid y) + g^\top (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\top P (\theta - \theta_0) + O(\|\theta - \theta_0\|^3), \quad (1)$$

where

$$g := \nabla \log p(\theta_0 \mid y) \quad \text{and} \quad P := -\nabla^2 \log p(\theta_0 \mid y).$$

Motivated by (1), we define our approximation to the log posterior $f_1 : \mathcal{X}_1 \rightarrow \mathbb{R}$ by

$$f_1(\theta) := \log p(\theta_0 \mid y) + g^\top (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\top P (\theta - \theta_0).$$

2. We have that

$$g = \nabla \log p(\theta_0 | y) = \nabla (\log p(y | \theta_0) + \log p(\theta_0) - \log p(y)) = \nabla \log p(y | \theta_0) + \nabla \log p(\theta_0). \quad (2)$$

For $\theta \in \mathcal{X}_1$, we have that

$$\nabla \log p(y | \theta) = \nabla \left(-\frac{1}{2\sigma^2} \|y - A(\theta)\|_2^2 \right) = \frac{1}{\sigma^2} J_A(\theta)^\top (y - A(\theta)),$$

where $J_A(\theta)$ is the Jacobian of the forward map A (restricted to the domain $\mathcal{X}_1$). Recalling that $y = A(\theta_0)$,

$$\nabla \log p(y | \theta_0) = \frac{1}{\sigma^2} J_A(\theta_0)^\top (y - A(\theta_0)) = 0. \quad (3)$$

Finally, for $\theta \in \mathcal{X}_1$, we have that

$$\nabla \log p(\theta) = \nabla \log (\lambda_R e^{-\lambda_R R}) = \nabla (-\lambda_R R) = (0, 0, -\lambda_R, 0)^\top. \quad (4)$$

Combining (2), (3), and (4), we have that

$$g = (0, 0, -\lambda_R, 0)^\top.$$

3. The negative Hessian of the log posterior is given by

$$P = -\nabla^2 \log p(\theta_0 | y) = -\nabla^2 \log p(y | \theta_0) - \nabla^2 \log p(\theta_0). \quad (5)$$

Recalling (4), we have that

$$\nabla^2 \log p(\theta_0) = 0. \quad (6)$$

Next, for $\theta \in \mathcal{X}_1$, one can show that

$$\nabla^2 \log p(y | \theta) = -\frac{1}{\sigma^2} J_A(\theta)^\top J_A(\theta) + \frac{1}{\sigma^2} \sum_k (y - A(\theta))_k \nabla^2 A_k(\theta).$$

Again recalling that $y = A(\theta_0)$, we obtain

$$\nabla^2 \log p(y | \theta_0) = -\frac{1}{\sigma^2} J_A(\theta_0)^\top J_A(\theta_0). \quad (7)$$

Combining (5), (6), and (7), we have that

$$P = \frac{1}{\sigma^2} J_A(\theta_0)^\top J_A(\theta_0). \quad (8)$$

To obtain a formula for P , we will obtain a formula for $J_A(\theta_0)$. To this end, recall that the $tN_x + x$ -th entry of the forward map $A : \mathcal{X} \rightarrow [0, 1]^{N_t N_x}$ with parameters $\theta = (x_0, d, R, a)^\top \in \mathcal{X}_1$ is given by

$$A(t, x; \theta) = aB(x)w(t - T(x))$$

where

$$B(x) = \sqrt{\frac{d+R}{s(x)}} e^{-2\alpha s(x)}, \quad T(x) = \frac{2}{v}(s(x) - (d+R)) + \frac{2d}{v} = \frac{2}{v}(s(x) - R)$$

with $s(x) = \sqrt{(d+R)^2 + (x - x_0)^2}$, and

$$w(\tau) = (1 - 2(\pi f_0 \tau)^2) e^{-(\pi f_0 \tau)^2}.$$

Applying the product and chain rules, we obtain

$$\begin{aligned}
\frac{\partial A(t, x; \theta)}{\partial x_0} &= aB(x) \left[w(t - T(x)) \frac{\partial \log B(x)}{\partial x_0} + w'(t - T(x)) \frac{-\partial T(x)}{\partial x_0} \right], \\
\frac{\partial A(t, x; \theta)}{\partial d} &= aB(x) \left[w(t - T(x)) \frac{\partial \log B(x)}{\partial (d + R)} + w'(t - T(x)) \frac{-\partial T(x)}{\partial d} \right], \\
\frac{\partial A(t, x; \theta)}{\partial R} &= aB(x) \left[w(t - T(x)) \frac{\partial \log B(x)}{\partial (d + R)} + w'(t - T(x)) \frac{-\partial T(x)}{\partial R} \right], \\
\frac{\partial A(t, x; \theta)}{\partial a} &= B(x) w(t - T(x)).
\end{aligned} \tag{9}$$

The needed derivatives are given by

$$\begin{aligned}
w'(\tau) &= 2(\pi f_0)^2 \tau (2(\pi f_0 \tau)^2 - 3) e^{-(\pi f_0 \tau)^2}, \\
\frac{\partial \log B(x)}{\partial x_0} &= (x - x_0) \left(\frac{1}{2s(x)^2} + \frac{2\alpha}{s(x)} \right), \\
\frac{\partial \log B(x)}{\partial (d + R)} &= \frac{1}{2(d + R)} - \frac{(d + R)}{2s(x)^2} - \frac{2\alpha(d + R)}{s(x)}, \\
-\frac{\partial T(x)}{\partial x_0} &= \frac{2(x - x_0)}{vs(x)}, \\
-\frac{\partial T(x)}{\partial d} &= -\frac{2(d + R)}{vs(x)}, \\
-\frac{\partial T(x)}{\partial R} &= \frac{2}{v} \left(1 - \frac{(d + R)}{s(x)} \right).
\end{aligned}$$

Vectorising each of the derivatives in (9) gives the four columns of $J_A(\theta_0)$.

4. Recall that for $\theta \in \mathcal{X}_1$, our approximation to the log posterior is given by

$$f_1(\theta) = \log p(\theta_0 | y) + g^\top (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\top P (\theta - \theta_0).$$

Assuming that P is positive definite and so P^{-1} exists (which holds if and only if $J_A(\theta_0)$ has full rank), we may complete the square to obtain

$$\begin{aligned}
f_1(\theta) &= \log p(\theta_0 | y) + g^\top (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\top P (\theta - \theta_0) \\
&= \log p(\theta_0 | y) - \frac{1}{2} [(\theta - \theta_0)^\top P (\theta - \theta_0) - 2g^\top (\theta - \theta_0)] \\
&= \log p(\theta_0 | y) - \frac{1}{2} [(\theta - \theta_0)^\top P (\theta - \theta_0) - 2g^\top (\theta - \theta_0) + g^\top P^{-1}g - g^\top P^{-1}g] \\
&= \log p(\theta_0 | y) - \frac{1}{2} [(\theta - \theta_0 - P^{-1}g)^\top P (\theta - \theta_0 - P^{-1}g) - g^\top P^{-1}g] \\
&= \log p(\theta_0 | y) + \frac{1}{2} g^\top P^{-1}g - \frac{1}{2} (\theta - \theta_0 - P^{-1}g)^\top P (\theta - \theta_0 - P^{-1}g) \\
&= C' - \frac{1}{2} (\theta - \mu)^\top P (\theta - \mu),
\end{aligned}$$

where C' is some constant independent of θ , and $\mu := \theta_0 + P^{-1}g$.

Thus, our approximation $f_1(\theta)$ defines a truncated Gaussian on $\mathcal{X}_1$, with mean μ and covariance P^{-1} . In the implementation, the Moore–Penrose pseudoinverse P^+ is used in place of P^{-1} for numerical robustness.

Question 2.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide a function that approximates the log likelihood $\log p(y \mid \theta)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $g_2 : \mathcal{X}_2 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $g_2(\theta_1, \theta_2) = g_2(\theta_2, \theta_1)$;
3. a good approximation to the log likelihood in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. **TODO: read over.** For $(\theta_1, \theta_2) \in \mathcal{X}_2$, the log likelihood is given by

$$\log p(y \mid \theta_1, \theta_2) = -\frac{1}{2\sigma^2} \|A(\theta_0) - A(\theta_1) - A(\theta_2)\|_2^2 + C$$

where C is some constant independent of (θ_1, θ_2) , and $A : \mathcal{X} \rightarrow [0, 1]^{N_t N_x}$ is the forward map. Recall that the $tN_x + x$ -th entry of the forward map $A : \mathcal{X} \rightarrow [0, 1]^{N_t N_x}$ with parameters $\theta = (x_0, d, R, a)^\top \in \mathcal{X}_1$ is given by

$$A(t, x; \theta) = aB(x)w(t - T(x))$$

where

$$B(x) = \sqrt{\frac{d+R}{s(x)}} e^{-2\alpha s(x)}, \quad T(x) = \frac{2}{v}(s(x) - (d+R)) + \frac{2d}{v} = \frac{2}{v}(s(x) - R)$$

with $s(x) = \sqrt{(d+R)^2 + (x - x_0)^2}$, and

$$w(\tau) = (1 - 2(\pi f_0 \tau)^2) e^{-(\pi f_0 \tau)^2}.$$

Reparameterization Define $g_0 := (x_0, d_0, R_0)^\top$, so that we may write the representative atom as

$$\theta_0 = (g_0, a_0)^\top.$$

Now, with two atoms

$$\theta_1 = (g_1, a_1)^\top \quad \text{and} \quad \theta_2 = (g_2, a_2)^\top,$$

where $g_1 := (x_1, d_1, R_1)^\top$ and $g_2 := (x_2, d_2, R_2)^\top$, we will use the following reparameterization:

- $m := a_1 + a_2$ is the total amplitude,
- $s := \frac{a_1}{a_1 + a_2}$ is the amplitude split,
- $c := sg_1 + (1 - s)g_2$ is the amplitude-weighted center,
- $h := g_1 - g_2$ is the difference.

We may recover the original parameters as follows:

- $a_1 = sm$,
- $a_2 = (1 - s)m$,
- $g_1 = c + (1 - s)h$,
- $g_2 = c - sh$.

Rewrite the forward map Define the function $F : [0, X_{\max}] \times [0, D_{\max}] \times [0, \infty) \rightarrow [0, 1]^{N_t N_x}$ by

$$F(g) := A(g, 1),$$

where $A(g, 1)$ is the $N_t N_x$ -dimensional vector with $A(t, x; g, 1)$ as the $tN_x + x$ -th entry. Notice that we have that

$$A(g, a) = aF(g).$$

Hence,

$$\begin{aligned} A(\theta_1) + A(\theta_2) &= a_1 F(g_1) + a_2 F(g_2) \\ &= m [sF(g_1) + (1 - s)F(g_2)] \\ &= m [sF(c + \delta_1) + (1 - s)F(c + \delta_2)]. \end{aligned} \quad (10)$$

where $\delta_1 := g_1 - c = (1 - s)h$ and $\delta_2 := g_2 - c = -sh$.

Taylor approximation to F around c Now, (since F is smooth) performing a second-order Taylor expansion of F around c gives

$$F(c + \delta) = F(c) + J_c \delta + \frac{1}{2} H_c[\delta, \delta] + O(\|\delta\|^3), \quad (11)$$

where J_c is the Jacobian of F at c , given by

$$J_c = \left[\frac{\partial F(c)}{\partial x_0}, \frac{\partial F(c)}{\partial d}, \frac{\partial F(c)}{\partial R} \right]$$

and $H_c[\delta, \delta]$ is the $N_t N_x$ -dimensional vector with k -th entry

$$(H_c[\delta, \delta])_k = \delta^T \nabla^2 F_k(c) \delta,$$

where $F_k(c)$ is the k -th entry of $F(c)$.

Using the Taylor approximation (11) for each term in (10) gives

$$\begin{aligned} sF(c + \delta_1) + (1 - s)F(c + \delta_2) &= F(c) + J_c (s\delta_1 + (1 - s)\delta_2) + \frac{1}{2} (sH_c[\delta_1, \delta_1] + (1 - s)H_c[\delta_2, \delta_2]) \\ &\quad + O(\|\delta_1\|^3 + \|\delta_2\|^3). \end{aligned} \quad (12)$$

Now, notice that since $s\delta_1 + (1 - s)\delta_2 = 0$, we have that

$$J_c (s\delta_1 + (1 - s)\delta_2) = 0. \quad (13)$$

Next, notice that since $H_c[\delta, \delta]$ is bilinear, we have that

$$\begin{aligned} sH_c[\delta_1, \delta_1] + (1 - s)H_c[\delta_2, \delta_2] &= s(1 - s)^2 H_c[h, h] + (1 - s)s^2 H_c[h, h] \\ &= s(1 - s)H_c[h, h]. \end{aligned} \quad (14)$$

Combining (12), (13), and (14) gives

$$sF(c + \delta_1) + (1 - s)F(c + \delta_2) = F(c) + \frac{1}{2} s(1 - s)H_c[h, h] + O(\|\delta_1\|^3 + \|\delta_2\|^3). \quad (15)$$

Since

$$\|\delta_1\|^3 + \|\delta_2\|^3 = (|s|^3 + |1-s|^3) \|h\|^3,$$

if we restrict to the region where $|a_1 + a_2| > \epsilon$ for some $\epsilon > 0$ (which should be reasonable otherwise they cancel each other out in the image), so that $|s|, |1-s| < 1/\epsilon$, then we have that

$$O(\|\delta_1\|^3 + \|\delta_2\|^3) = O(\|h\|^3). \quad (16)$$

Overall, combining (15) and (16) gives

$$sF(c + \delta_1) + (1-s)F(c + \delta_2) = F(c) + \frac{1}{2}s(1-s)H_c[h, h] + O(\|h\|^3) \quad (17)$$

when $|a_1 + a_2| > \epsilon$.

Taylor approximation to F around g_0 Now, (since F is smooth) performing a first-order Taylor expansion of F around g_0 gives

$$F(g_0 + \delta) = F(g_0) + J_{g_0}\delta + O(\|\delta\|^2), \quad (18)$$

where and J_{g_0} is the Jacobian of F at g_0 , given by

$$J_{g_0} = \left[\frac{\partial F(g_0)}{\partial x_0}, \frac{\partial F(g_0)}{\partial d}, \frac{\partial F(g_0)}{\partial R} \right].$$

Applying the approximation (18) to $F(c)$ in (17) gives

$$\begin{aligned} sF(c + \delta_1) + (1-s)F(c + \delta_2) &= F(c) + \frac{1}{2}s(1-s)H_c[h, h] + O(\|h\|^3) \\ &= F(g_0 + q) + \frac{1}{2}s(1-s)H_c[h, h] + O(\|h\|^3) \\ &= F(g_0) + J_{g_0}q + \frac{1}{2}s(1-s)H_c[h, h] + O(\|q\|^2 + \|h\|^3). \end{aligned} \quad (19)$$

when $|a_1 + a_2| > \epsilon$, where $q := c - g_0$.

Approximate $H_c[h, h]$ by $H_{g_0}[h, h]$ Since we would like an approximation that only depends on the representative atom (g_0, a_0) , we will approximate $H_c[h, h]$. Given some assumptions (such as F has bounded third derivatives in a neighbourhood of g_0), we have that

$$H_c[h, h] = H_{g_0}[h, h] + O(\|q\|\|h\|^2).$$

Combining this with (19) gives

$$\begin{aligned} sF(c + \delta_1) + (1-s)F(c + \delta_2) &= F(g_0) + J_{g_0}q + \frac{1}{2}s(1-s)H_{g_0}[h, h] \\ &\quad + O(\|q\|^2 + \|q\|\|h\|^2 + \|h\|^3), \end{aligned} \quad (20)$$

when $|a_1 + a_2| > \epsilon$.

Approximation of the forward map Define $\eta := m - a_0$ to be the difference between the total amplitude and the amplitude of the representative atom. Then, from (20) and (10) we have that

$$\begin{aligned} A(\theta_1) + A(\theta_2) &= m[sF(c + \delta_1) + (1-s)F(c + \delta_2)] \\ &= (a_0 + \eta)[sF(c + \delta_1) + (1-s)F(c + \delta_2)] \\ &= (a_0 + \eta)\left[F(g_0) + J_{g_0}q + \frac{1}{2}s(1-s)H_{g_0}[h, h] \right. \\ &\quad \left. + O(\|q\|^2 + \|q\|\|h\|^2 + \|h\|^3)\right] \\ &= a_0F(g_0) + \eta F(g_0) + a_0J_{g_0}q + \frac{a_0}{2}s(1-s)H_{g_0}[h, h] \end{aligned}$$

$$\begin{aligned}
& + O(|\eta| \|q\| + |\eta| \|h\|^2 + \|q\|^2 + \|q\| \|h\|^2 + \|h\|^3) \\
& = A(\theta_0) + \eta F(g_0) + a_0 J_{g_0} q + \frac{a_0}{2} s(1-s) H_{g_0}[h, h] \\
& + O(|\eta| \|q\| + |\eta| \|h\|^2 + \|q\|^2 + \|q\| \|h\|^2 + \|h\|^3)
\end{aligned}$$

when $|a_1 + a_2| > \epsilon$. That is,

$$\begin{aligned}
A(\theta_1) + A(\theta_2) - A(\theta_0) & = \eta F(g_0) + a_0 J_{g_0} q + \frac{a_0}{2} s(1-s) H_{g_0}[h, h] \\
& + O(|\eta| \|q\| + |\eta| \|h\|^2 + \|q\|^2 + \|q\| \|h\|^2 + \|h\|^3)
\end{aligned} \tag{21}$$

in some neighborhood around $\eta = m - a_0 = 0$.

Question 3.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_2 : \mathcal{X}_2 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $f_2(\theta_1, \theta_2) = f_2(\theta_2, \theta_1)$;
3. a good approximation to the log posterior in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant. This should build on the function found in Question 2.

Solution.

Question 4.**[pts.]**

Fix $n \geq 2$. Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_n$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_n : \mathcal{X}_n \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable;
3. a good approximation to the log posterior in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant, and this constant need not be common to all n . This should be a generalization of the function found in Question 3.

Solution.

Question 5.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Rewrite the function found in Question 1 in the same form as the functions found in Question 4.

If it is not possible to write the function found in Question 1 in the same form as the functions found in Question 4, explain why. Then, provide an alternative function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_1$ that is in the same form as the functions found in Question 4.

Solution.

Question 6.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide an approximation of the sequence of probabilities $\{p_n\}_{n \in \mathbb{N}}$ where

$$p_n := P(\theta \in X_n \mid y) = \int_{\mathcal{X}_n} p(\theta \mid y) d\theta,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0 .

You may find it useful to use the functions found in Questions 4 and 5. Note that you need only approximate the sequence up to a common multiplicative constant; that is, it suffices to approximate quantities proportional to $\{p_n\}_{n \in \mathbb{N}}$.

Solution.

Question 7.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide an approximation of the probability p where

$$p :=,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0 .

Solution.